

SECTION B

7 (**short**) The arm of a rotary mixer consists of four semi-circular plates attached to a thin light axle, as shown in Fig. 4. Each plate has mass M , diameter $2R$, and the mass and diameter of the axle and the thickness of the plates are negligible.

- (a) Using an axis theorem, calculate the moment of inertia of the mixer arm around its axis. [5]
- (b) If $M = 100$ g, and $R = 10$ cm, determine the average power required to spin up the mixer arm in vacuum in 1 second from rest to 600 rpm. [5]

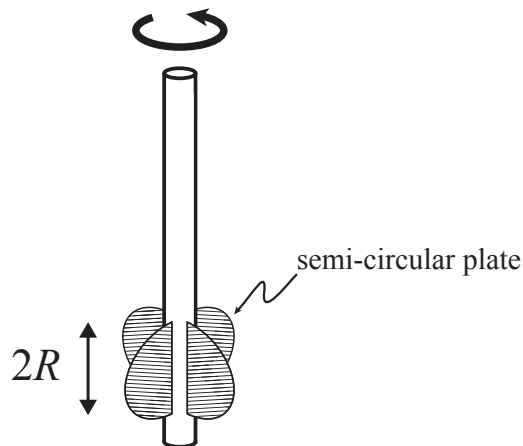


Fig. 4